INCH POUND

MIL-PRF-46846D
02May2013
SUPERSEDING
MIL-PRF-46846C
29December1998

PERFORMANCE SPECIFICATION

RUBBER, SYNTHETIC, HEAT - SHRINKABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the type and class requirements for extruded tubing, molded parts, and extruded shapes that are flexible, electrical insulating and whose dimensions will reduce to a predetermined size upon the application of heat.

1.2 Classification. The heat-shrinkable rubber will consist of the following types and classes, as specified (see 6.2).

Type I. Cross-linked, modified polychloroprene (Neoprene) molded parts, and extruded shapes, flexible general purpose cable-jacketing, cable harness, boots and transitions, service temperature range of -55 to 90°C.

Class 1 - Deleted (see 6.4.1).
Class 2 - Molded parts and extruded shapes, cable harness boots and transitions, expansion ratio 2.5.

Type II. Cross-linked, modified flame-retardant silicone rubber tubing, molded parts and extruded shapes, highly flexible, ablative-resistant.

Class 1 - Tubing, expansion ratio 1.75, service temperature range of -70 to 180°C.

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Class 2 - Molded parts and extruded shapes, expansion ratio 2 to 4, service temperature range -75 to 135°C.

Type III. Cross-linked, fluoroelastomer, modified polymer of hexafluoropropylene and vinylidene fluoride dipolymer, molded parts and extended periods of high temperature and solvent attack.

Class 1 - Tubing, expansion ratio 2, service temperature range -55 to 200°C.
Class 2 - Molded parts and extruded shapes, expansion ratio 3.5, service temperature range of -55 to 200°C.

Type IV. Cross-linked, modified, propellant-resistant butyl rubber tubing, molded parts and extruded shapes, flexible for protection of cable assemblies exposed to hypergolic fuels.

Class 1 - Tubing, expansion ratio 2, service temperature range of -55 to 125°C.
Class 2 - Molded parts and extruded shapes, expansion ratio 2.5, service temperature range of -55 to 125°C.

Type V. Deleted (see 6.4.2).

Type VI. Cross-linked, modified, flame-resistant ethylene-propylene rubber tubing, molded parts, extruded shapes, flexible, ozone resistant, and mechanically tough.

Class 1 - Tubing, expansion ratio 2, service temperature of -70 to 115°C.
Class 2 - Molded parts and extruded shapes, expansion ratio 4, service temperature range of -75 to 125°C.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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STANDARDS FEDERAL

FED-STD-376 Preferred Metric Units for General Use By The Federal Government

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-5606 Hydraulic Fluid, Petroleum Base: Aircraft; Missile, and Ordnance
MIL-PRF-7254 Propellant, Nitric Acid
MIL-PRF-7808 Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-PRF-26539 Propellants, Dinitrogen Tetroxide
MIL-PRF-27402 Propellant, Hydrazine - UNS-Dimethylhydrazine (50% N_2H_4 - 50% UDMH)

(Copies of these documents are available online at http://assist.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D257 - Standard Test Method for DC Resistance or Conductance of Insulating Materials
ASTM D412 - Standard Test Method for Vulcanized Rubber and Thermoplastic Elastomers - Tension
ASTM D570 - Standard Test Method for Water Absorption of Plastics
ASTM D635 - Standard Test Method for Rate of Burning and/or Extent and Time of Burning of Plastics in a Horizontal Position
ASTM D746 - Standard Test Method for Brittleness Temperature of Plastics and Elastomers by Impact
ASTM D792 - Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement
ASTM D876 - Standard Test Method for Nonrigid Vinyl Chloride Polymer Tubing Used for Electrical Insulation
ASTM D1149 - Standard Test Method for Rubber Deterioration Surface Ozone Cracking in a Chamber
ASTM D2240 - Standard Test Method for Rubber Property-Durometer Hardness
ASTM D2671 - Standard Test Method for Heat-Shrinkable Tubing for Electrical Use
ASTM G21 - Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi,

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(Copies of these documents are available from www.astm.org or ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Toxic chemicals, hazardous substances, and ozone depleting substances (ODS). The use of toxic chemicals, hazardous substances, or ODS shall be avoided, whenever feasible.

3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Materials. Type I synthetic rubber specified herein shall be fabricated from a cross-linked, stabilized, modified polychloroprene (Neoprene) composition. Type II synthetic rubber shall be fabricated from a cross-linked, stabilized, flame-retardant, modified silicone rubber composition. Type III synthetic rubber shall be fabricated from a cross-linked, stabilized, flame-resistant, fluorelastomer, modified polymer of hexafluoropropylene and vinylidene fluoride dipolymer. Type IV synthetic rubber shall be fabricated from a cross-linked stabilized, propellant-resistant, modified butyl rubber composition. Type VI synthetic rubber shall be fabricated from a cross-linked, stabilized, flame-resistant, modified ethylene-proplylene rubber composition. These expanded synthetic rubbers shall shrink to predetermined dimensions upon the application of heat as follows:

a. Types I, IV, and VI - Heat in excess of 135°C.
b. Types II, III - Heat in excess of 175°C.

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3.5 Dimensions and tolerances (see table I).

3.5.1 Extruded tubing.

3.5.1.1 Sizes and tolerances. Unless otherwise specified (see 6.2), types II through VI extruded tubing shall have the sizes and tolerances specified in table I, as supplied and after heating (see 4.4.2.3.3).

3.5.1.2 Length. The length of tubing shall be as specified in the contract or order (see 6.2).

3.5.2 Molded parts and extruded shapes. The dimensions and tolerances of heat-shrinkable molded parts and extruded shapes shall be as specified on the applicable drawings or as specified in the contract or order (see 6.2).

3.5.3 Nonstandard dimensions.

3.5.3.1 Nonstandard thicknesses and shapes. Nonstandard thicknesses and shapes shall be specified by the procuring activity. Dimensional tolerances for these nonstandard thicknesses or shapes shall conform to the tolerances for the nearest standard size, unless otherwise specified (see 6.2).

3.5.3.2 Length. If applicable, the length of tubing shall be as specified in the contract or order (see 6.2).

3.5.4 Storage. The heat-shrinkable tubing, molded parts, and extruded shapes shall meet the requirements of this specification after accelerated storage conditioning (see 4.3.1 and 4.3.2).

3.6 Physical, electrical, and chemical properties.

3.6.1 Tubing. Types II through VI, class 1, synthetic rubber tubing shall have the physical, electrical, and chemical property requirements as specified in table II (see 4.4.2.1).

3.6.2 Molded parts and extruded shapes. Types I through VI, class 2, synthetic rubber molded parts and extruded shapes, shall have the physical, electrical, and chemical property requirements (see 4.4.2.2) as specified in table III.

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TABLE I. Tubing sizes and tolerances, types II through VI, class 1.

EXPANDED - AS SUPPLIED					RECOVERED DIMENSIONS - AFTER HEATING							
MINIMUM INSIDE DIAMETER, INCHES					MAXIMUM INSIDE DIAMETER, INCHES				WALL THICKNESS, INCH			
Size, Inches	Type II	Type III	Type IV	Type VI	Type II	Type III	Type IV	Type VI	Type II	Type III	Type IV	Type VI
0.25	0.250	0.250	0.250	0.250	0.143	0.125	0.125	0.125	.035±.010	.030±.010	.035±.010	.030±.010
0.375	0.375	0.375	0.375	0.375	0.214	0.187	0.187	0.187	.040±.010	.035±.010	.040±.010	.035±.010
0.5	0.500	0.500	0.500	0.500	0.286	0.250	0.250	0.250	.048±.010	.043±.010	.050±.015	.043±.010
0.625	0.625	0.625	---	0.625	0.357	0.312	---	0.312	.052±.015	.047±.012	---	.047±.012
0.75	0.750	0.750	0.750	0.750	0.428	0.375	0.375	0.375	.057±.015	.052±.015	.060±.015	.052±.015
0.875	0.875	0.875	---	0.875	0.500	0.437	---	0.437	.065±.015	.060±.015	---	.060±.015
1.0	1.00	1.00	1.00	1.00	0.570	0.500	0.500	0.500	.070±.015	.065±.015	.075±.020	.065±.015
1.25	1.25	1.25	---	1.25	0.714	0.625	---	0.625	.087±.020	.070±.020	---	.070±.020
1.5	1.50	1.50	1.50	1.50	0.857	0.750	0.750	0.750	.095±.020	.075±.020	.100±.020	.075±.020
1.75	1.75	---	---	---	1.00	---	---	---	.107±.020	---	---	---
2.0	2.00	2.00	2.00	---	1.14	1.00	1.00	---	.110±.020	.110±.020	.110±.020	---
2.5	---	---	2.50	---	---	---	1.25	---	---	---	.130±.020	---
3.0	---	---	3.00	---	---	---	1.50	---	---	---	.140±.020	---
4.0	---	---	4.00	---	---	---	2.00	---	---	---	.150±.020	---

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TABLE II. Physical, electrical, and chemical properties of synthetic rubber tubing, class 1.

Property	Conditioning (see 4.4.2.3)				Requirement				Method
	Type II	Type III	Type IV	Type VI	Type II	Type III	Type IV	Type VI	
Longitudinal Change percent, maximum	A	A	A	A	10	20	15	15	Paragraph 4.5.3
Tensile Strength, psi, minimum					600	1200	1200	1400	(See 4.4.2.1)
Tensile Stress at 100 percent elongation, psi, maximum					500	1400	1200	1000	(See 4.4.2.1)
Elongation, percent, minimum					200	250	250	200	(See 4.4.2.1)
Specific Gravity, maximum					1.35	2.0	1.40	1.25	(See 4.4.2.1)
Hardness, Shore A	A	A	A	A	60 ± 5	82 ± 5	85 ± 5	80 ± 5	ASTM D2240
Low Temperature	E4/-95	E4/-40	E4/-67	E4/-103	Pass or Fail	Pass or Fail	Pass or Fail	Pass or Fail	(See 4.4.2.1)
Heat Shock	E4/572	E4/-572	E4/392	E4/392	←---------- No dripping, flowing or cracking----------→				Paragraph 4.5.4
Heat Aging, followed by tests for:									Paragraph 4.5.5
Tensile Strength, psi, minimum	E168/352	E168/482	E96/902	E168/302	500	1200	1000	1200	(See 4.4.2.1)
Elongation, percent, minimum	E168/392	E168/482	E168/302	E96/302	120	200	100	100	(See 4.4.2.1)
Dielectric Strength, volts/mil, minimum	A	A	A	A	300	150	100	220	(See 4.4.2.1)
Volume Resistivity, ohm-cm, minimum					1×10^{11}	1×10^{9}	1×10^{10}	1×10^{11}	(See 4.4.2.1)
Flammability, seconds maximum					60: self-exting. 1/ Method A	15: self exting. Method B	---	15: self exting. Method A	(See 4.4.2.1) (Use indicated method)
Fungus Resistance	A	A	A	A	<---------- Rating of 0 ----------→				ASTM G21
Water Absorption, percent, maximum	E24/77	E24/77	E24/77	E24/77	1.0	0.5	0.5	1.0	(See 4.4.2.1) (Procedure A)
Solvent Resistance, followed by tests for:									Paragraphs 3.7 and 4.5.6
Tensile strength, psi, minimum	E24/77	E24/77	E1/2/77	E24/77	400	1200	1000	500	(See 4.4.2.1)
Elongation, percent, minimum	E24/77	E24/77	E1/2/77	E24/77	120	250	200	100	(See 4.4.2.1)
Eccentricity, percent, minimum	A	A	A	A	60	60	60	60	ASTM D2671

1/ Not more than 25 percent of indicator flag burned; no dripping or flowing.

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TABLE III. Physical, electrical, and chemical properties of synthetic rubber molded parts and extruded shapes, class 2.

Property	Conditioning (see 4.4.2.3)					Requirement					Method
	Type I	Type II	Type III	Type IV	Type VI	Type I	Type II	Type III	Type IV	Type VI	
Elastic Memory, percent, minimum	A	A	A	A	A	Expansion 150 Retraction 90	Expansion 250 Retraction 90	Expansion 200 Retraction 90	Expansion 150 Retraction 90	Expansion 275 Retraction 93	Paragraph 4.5.2
Tensile Strength, psi, minimum						1500	600	1800	1200	1600	ASTM D412, Die D
Tensile Stress at 100 percent elongation, psi, maximum						750	600	1200	1000	750	ASTM D412, Die D
Elongation, percent, minimum						200	200	300	250	450	ASTM D412, Die D
Specific Gravity, maximum						1.50	1.35	1.95	1.40	1.25	ASTM D792
Hardness, Shore A	A	A	A	A	A	78 ± 5	83 ± 5	85 ± 5	90 ± 5	85 ± 5	ASTM D2240
Low Temperature	E4/-67	E4/-104	E4/-87	E4/-67	E4/-103	Pass or Fail	Pass or Fail	Pass or Fail	Pass or Fail	Pass or Fail	ASTM D746
Heat Shock	E4/302	E4/432	E4/572	E4/572	E4/437	<---------- No dripping, flowing, or cracking ---------->					Paragraph 4.5.4
Heat Aging, followed by tests for:											Paragraph 4.5.5
Tensile Strength, psi, minimum	E96/212	E96/347	E168/482	E96/302	E168/302	1200	480	1200	1000	1500	ASTM D412, Die D
Elongation, percent, minimum	E96/212	E96/347	E168/482	E96/302	E168/302	150	120	250	100	150	ASTM D412, Die D
Dielectric Strength, volts/mil, minimum	A	A	A	A	A	200	200	200	100	225	ASTM D876
Volume Resistivity, ohm-cm, minimum						1×10^{8}	1×10^{10}	1×10^{10}	1×10^{10}	1×10^{11}	ASTM D257
Flammability						<------ Self-Extinguishing ------>			---	Self-Extinguishing	ASTM D635
Fungus Resistance	A	A	A	A	A	<---------- Rating of 0 ---------->					ASTM G21
Water Absorption, percent, maximum	E24/77	E24/77	E24/77	E24/77	E24/77	1.0	1.0	0.5	0.5	1.0	ASTM D570, Procedure A
Solvent Resistance, followed by tests for:											Paragraph 4.5.6
Tensile strength, psi, minimum	E24/77	E24/77	E24/77	E1/77	E24/77	900	400	1500	1200	600	ASTM D412, Die D
Elongation, percent, minimum	E24/77	E24/77	E24/77	E1/77	E24/77	120	120	300	250	200	

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3.7 Solvent resistance. Type I class 2, type II classes 1 and 2, type III classes 1 and 2, type IV classes 1 and 2, and type VI classes 1 and 2, shall be resistant to solvents when tested in accordance with tables II and III (see 4.5.6).

3.8 Propellant resistance, type IV. Type IV classes 1 and 2 shall be resistant to hydrazine/unsymmetrical dimethylhydrazine (UDMH), nitrogen tetroxide, and nitric acid (see 4.5.7).

WARNING

UDMH is hazardous material and should be handled in accordance with the manufacturer's material safety data sheet.

3.9 Eccentricity of tubing. A thin wall area, due to eccentricity, shall be not greater than that shown in table II (see 4.5.8).

3.10 Ozone resistance type VI. When specified in the contract or order (see 6.2), type VI classes 1 and 2 shall be tested for ozone resistance requirements (see 4.5.9).

3.11 Color. All types and classes of the tubing, molded parts and extruded shapes, shall be black (see 4.5.1).

3.12 Workmanship. The synthetic rubber tubes, parts, and shapes shall be free from foreign material, flaws, pinholes, bubbles, cracks, inclusions, and excessive flash, and shall be homogeneous in texture (see 4.4.1.1).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed on a lot. A lot shall consist of that quantity of synthetic rubber compound produced in one continuous operation by the same manufacturer and manufactured in accordance with the same specification and specification revision.

4.3 First article inspection. First article inspection shall consist of all the examinations and tests specified herein plus the storage tests (see 4.3.1 and 4.3.2).

4.3.1 Low-temperature storage. The synthetic rubber shall be placed in a temperature chamber at a temperature of -55 $\pm$ 30°C for not less than 72 hours. Upon completion of the

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72-hour exposure, the synthetic rubber shall be allowed to stabilize at normal room temperature and shall then be subjected to all of the conformance tests specified herein (see 4.4).

4.3.2 Ambient storage. The synthetic rubber shall be placed in a temperature chamber at a temperature of 40 ± 3°C for not less than 336 hours. Upon completion of the 336-hour exposure, the synthetic rubber shall be allowed to stabilize at normal room temperature and shall then be subjected to tests for dimensions, as supplied and after heating in accordance with 4.5.3 (see 4.4.1 for sampling).

4.4 Conformance inspection.

4.4.1 Sampling.

4.4.1.1 Nondestructive examinations. Nondestructive examinations shall consist of examinations for the workmanship requirements of 3.12.

4.4.1.2 Destructive examinations and tests. Unless otherwise specified (see 6.2), samples for destructive examinations and tests of tubing shall be selected at random from the lot (see 4.4.2.1). The samples for destructive examinations and tests of molded parts shall be cut from molded slabs, 6 inches x 6 inches x 0.075 ± 0.010 inches, fabricated from the same batch of material and subjected to the same degree of cross-linking as the molded parts. Destructive examinations and tests shall consist of examinations and tests for tensile strength, heat shock, hardness, elastic memory, and the dimensions and tolerance requirements of 3.5 (see 4.5.2 through 4.5.6).

4.4.2 Test methods.

4.4.2.1 Tests for type II through VI, class 1. Unless otherwise specified in table II, tests shall be performed in accordance with ASTM D2671 to determine tubing properties as required by 3.6.1. Unless otherwise specified, the specimens shall be full sections of tubing for sizes 0.375 and smaller. For sizes 0.5 and larger, specimens shall be cut in accordance with 'Die D' of ASTM D412.

4.4.2.2 Tests for type I through VI, class 2. Tests shall be conducted in accordance with test methods specified in table III to determine properties as required by 3.6.2.

4.4.2.3 Conditioning. Prior to testing, specimens shall be conditioned as specified in table II or III.

4.4.2.3.1 Designation. The type of conditioning required shall be designated as follows:

Condition A: As received; no special conditioning after shrinking in accordance with 4.5.3.
Condition E: Temperature conditioning.

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4.4.2.3.2 Conditioning procedure. The conditioning procedures required, with the exception of Condition A, shall be indicated by the following combinations of symbols. The numbers shall be separated from each other by a slant mark.

a. A capital letter indicating the type of conditioning.

b. A number indicating the duration of the conditioning in hours.

c. A number indicating the conditioning temperature in degrees Celsius.

4.4.2.3.3 Conditioning tolerance. Unless otherwise specified (see 6.2), the conditioning temperature tolerance and additional testing information shall be as specified in table IV.

4.5 Tests. Testing to determine compliance with the requirements of tables II and III shall be in accordance with the methods specified in the tables and in accordance with the paragraphs of this specification called out in tables II and III.

4.5.1 Examination of product. The heat-shrinkable tubing, molded parts and extruded shapes shall be carefully examined to determine conformance to all requirements for which a specific test method is not specified herein (see 3.11).

TABLE IV. Conditioning and testing information.

Conditioning procedure	Conditioning time	Tolerance temperature	Testing temperature
E (high temperature)	± 1 hour	± 2°C	23 ± 2°C
E (low temperature)	+ 30 minutes or - zero	± 2°C	-55 ± 2°C

4.5.2 Elastic memory. A 6 inch by 0.125 inch specimen (see 4.4.1.2) shall be marked with two parallel gauge lines 1 inch apart in the central portion of the specimen. The 2-inch portion of the specimen including both gauge lines shall then be heated for 1 minute at 150 ± 2°C for types I, II, IV, and VI class 2, for 5 minutes at 200 ± 5°C for type III class 2, removed from the heat and stretched within 10 seconds until the gauge lines are 3 inches apart for types I and IV class 2, 4 inches apart for type III class 2, or 5 inches apart for type VI class 2. The extended specimen shall then be cooled in water for 1 minute at 20 to 30°C, released from tension, removed from the water, and wiped dry. After 24 hours at room temperature, the distance between the gauge lines shall be measured and recorded as the extended length. The portion of the specimen including both gauge lines shall then be reheated for 1 minute at the temperature specified above, removed from the heat and allowed to retract. The specimen again shall be cooled for 1 minute in water at 20 to 30°C, removed from the water, and wiped dry. The distance between gauge lines

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shall then be measured and recorded as the retracted length. Expansion and retraction shall be calculated as follows:

$$\text{Expansion \%} = (\text{extended length} - 1) \times 100 \text{ percent}$$

$$\text{Retraction \%} = \frac{\text{extended length} - \text{retracted length}}{\text{percent extended length} - 1} \times 100$$

4.5.3 Dimensional recovery and longitudinal change. Five specimens (see 4.4.1.2) shall be measured for dimensions as supplied, conditioned for 10 minutes in an oven at a temperature of 175 ± 5°C for type I class 2, types II, IV, and VI classes 1 and 2, and 200 ± 5°C for type III classes 1 and 2, cooled to room temperature and then remeasured. Tubing specimens shall be measured for length ± 0.03125 before and after conditioning, and longitudinal change shall be calculated as a percentage of the original length (see 3.5.1 and 4.4.2.3.3).

4.5.4 Heat shock. Three test specimens (see 4.4.1.2) of each lot shall be conditioned in an oven as specified in table II or III. The oven shall have an air velocity of 100 to 200 feet per minute past the specimens. While at the conditioning temperature and upon removal, the specimens shall be examined for evidence of dripping, flowing, or cracking.

4.5.5 Heat aging. Three specimens (see 4.4.1.2) shall be conditioned in an oven as specified in table II or III. The oven shall have an air velocity of 100 to 200 feet per minute past the specimens. After conditioning, the specimens shall be removed from the oven, cooled to room temperature, and tested as specified in table II or III.

4.5.6 Solvent resistance. The solvent resistance tests for the types and classes specified in 3.7 shall be performed by completely immersing three specimens (see 4.4.1.2) in each of the following solvents for 24 ± 2 hours at 25 ± 2°C. The volume of solvent shall be not less than 20 times that of the combined specimens.

a. JP-4 fuel

b. Hydraulic fluid conforming to MIL-PRF-5606

c. Lubricating oil conforming to MIL-PRF-7808

d. Aviation gasoline 100/130

e. Water (tap)

After immersion for the specified time, the specimens shall be lightly wiped and air-dried for 30 to 60 minutes at immersion temperature. The specimens shall then be examined for evidence of decomposition, swelling, or tackiness before being subjected to the tensile strength and elongation tests specified in table II or III.

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4.5.7 Propellant resistance. Type IV classes 1 and 2 shall have three 6-inch specimens, prepared and measured in accordance with ASTM D412. The specimens shall then be completely immersed in each of the following propellants for 30 ± 5 minutes at 25 ± 3°C. The volume of the propellant shall be not less than 20 times that of the specimens. After immersion, the specimens shall be rinsed in water, lightly wiped, and air dried for 30 to 60 minutes at room temperature. The specimens shall then be tested for tensile strength and elongation which shall be in accordance with the solvent resistance requirements of tables II and III, and 3.8.

a. Hydrazine/UDMH (MIL-PRF-27402)

b. Nitrogen tetroxide or Dinitrogen teroxide (MIL-P-26539)

c. Nitric acid (MIL-P-7254)

4.5.8 Eccentricity. Examinations for eccentricity of tubing shall be in accordance with ASTM D2671 and shall meet the requirements of table II and 3.9.

4.5.9 Ozone resistance. Type VI classes 1 and 2 only. The specimens shall be exposed for 7 days to an ozone controlled concentration of 50 ± 3 parts per hundred million (pphm) of air at 38 ± 1°C in accordance with ASTM D1149. After conditioning, no cracks shall be visible when the specimens are examined with a seven-power magnifier (see 3.10).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

6.1.1 General. The heat-reactive synthetic rubber materials covered by this specification are intended for use in a wide range of applications requiring a tough, flexible synthetic rubber that will reduce to predetermined dimensions upon application of heat and will maintain good

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physical and electrical properties after heat shrinkage. Some usages include identification and covering splices in cables, lugs, pipes; protecting feed-throughs; jacketing of assemblies; insulating pigtail connections; sleeves for wave guides and rectifier leads; and covers for resistors and diodes.

6.1.2 Military unique. The heat-reactive synthetic rubber materials covered by this specification have unique military requirements in that some types and classes must be resistant to chemicals such as hydrazine/UDMH (MIL-PRF-27402), Dinitrogen tetroxide (MIL-P-26539) and nitric acid (MIL-P-7254) that are used as propellants in missiles. All types must be resistant to fuels and oils such as JP-4 fuel, hydraulic fluid (MIL-PRF-5606), lubricating oil (MIL-PRF-7808) and aviation gasoline 100/130, that are used on military aircraft. All types must be functional over military temperature ranges, which exceeds that required in the commercial market.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Types and classes as required (see 1.2).

c. When first article is required (see 3.1 and 4.3).

d. Sizes and tolerances required, if different (see 3.5.1, 3.5.2, and 3.5.3).

e. Length if required (see 3.5.1.2 and 3.5.3.2).

f. When ozone resistance tests for type VI are required (see 3.10).

g. If destructive examinations and tests of tubing are not required (see 4.4.1.2).

h. Conditioning tolerance if different (see 4.4.2.3.3).

i. Packaging (see 5).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample (see 4.3) or a number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bid should provide that the Government reserves the right to waive the requirement for first article inspections to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish

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evidence with the bid that prior Government approval is presently appropriate for the pending contract (see 3.1).

6.4 Deleted types and classes.

6.4.1 Type I class 1. Type I class 1 has been deleted from this specification because it duplicates Society of Automotive Engineers (SAE) Aerospace Material Specification (AMS) 3623D, Tubing, Irradiated Polycholoroprene Eastomer, Electrical Insulation, Flexible, Heat-Shrinkable, 1.750 to 1 Shrink Ratio.

6.4.2 Type V. Type V synthetic rubber has been deleted from this specification because it duplicates the synthetic rubber specified by MIL-DTL-23053/5.

6.5 Engineering information.

6.5.1 Service temperature range. The synthetic rubber covered by this specification is suitable for continuous use throughout the temperature range as follows:

Type	Temperature
I, class 2	-55 to 90 ± 2°C
II, class 1	-70 to 180 ± 5°C
II, class 2	-75 to 135 ± 5°C
III, classes 1 and 2	-55 to 200 ± 5°C
IV, classes 1 and 2	-55 to 125 ± 3°C
VI, class 1	-70 to 115 ± 3°C
VI, class 2	-75 to 125 ± 3°C

6.5.2 Insulation resistance. The synthetic rubber covered by this specification is not suitable for use by itself as an insulation over exposed bare wires or components where high insulation resistance (100 megohms resistance when 500 volts are applied) is required.

6.5.3 Selection of appropriate size tubing. The important considerations in selecting the appropriate size tubing for a particular application are the outside diameter of the part being covered and the final wall thickness of the heat shrunk tubing. Maximum efficiency usually is obtained when the diameter of the part being covered is approximately midway between the minimum expanded and maximum recovered diameters of the tubing.

6.6 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

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6.7 Subject term (keyword listing).

Extruded
Hydrazine
Molded
parts Nitric
acid
Nitrogen tetroxide
Dinitrogen tetroxide
Shrink
tube
Tubing
UDMH
Unsymmetrical dimethylhydrazine

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
DLA - GS
Army –MI

Reviewers:
Army – CR4, MR

Preparing Activity:
DLA - GS8

(Project 5970 -2013-002)

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